

TN SKILL PROJECT

Eight-Phase Project Plan & Implementation Document

Project :Auto Ticket Classification Using Flow Designer

Platform	ServiceNow
Automation Tool	Flow Designer
Business Area	IT Service Management (ITSM) / Incident & Request Management
Core Workflow	Automated Ticket Ingestion → AI/Rule-Based Category & Priority Classification → Dynamic Group Assignment
Team	4 member including Team Lead

Project Overview

This project addresses operational delays and misclassification issues in IT Service Management ticket routing. Unclassified or misrouted tickets cause prolonged MTTR (Mean Time to Resolution), manual overhead for Service Desk agents, and poor SLA compliance.

The proposed solution utilizes **ServiceNow Flow Designer** to implement an automated ticket classification workflow. Upon ticket creation (Incident/Catalog Request), the flow evaluates ticket input attributes (short description, channel, category keywords), dynamically determines the correct Category, Subcategory, and Impact/Urgency levels, and automatically routes the ticket to the appropriate Assignment Group.

End-to-End Process

1. A user submits an IT Ticket/Incident via the Service Portal or Email Ingestion.
2. The ServiceNow Flow Designer trigger fires on new record creation.
3. The flow executes logic to analyze incoming ticket fields.
4. The flow assigns values to **Category**, **Subcategory**, **Impact**, and **Urgency**.
5. The **Assignment Group** is updated dynamically (e.g., Service Desk, Network, Software, Database).
6. The updated ticket is logged, and assignment notifications are dispatched automatically.

Phase 1: Brainstorming & Ideation

Purpose

Identify operational bottlenecks in IT ticket handling and convert them into an automated classification and routing workflow.

Problem Identified

Manual ticket triage requires dedicated Service Desk personnel to read, categorize, and route incoming incidents. This leads to:

- High rates of initial misclassification.
- Increased time-to-assignment and delayed response times.
- Bottlenecks during peak IT service outage periods.
- Inconsistent priority scoring across different Service Desk analysts.

User Story

As an IT Service Desk Leader, I want incoming support tickets to be automatically classified and assigned based on their content, so that resolution teams receive tickets immediately without manual triage intervention, reducing average resolution times and human error.

Proposed Idea

- Use **ServiceNow Flow Designer** triggered upon Incident Creation (incident table).
- Process ticket text attributes (Short Description, Category) using decision rules.
- Automatically update **Category, Subcategory, Priority, and Assignment Group**.
- Log automated work notes on the ticket for auditing purposes.

Expected Value

- **Reduced Triage Delay:** Instant automated assignment upon creation.
- **Lower Misrouting Rate:** Uniform decision logic applied to incoming tickets.
- **Improved SLA Compliance:** Quicker response start times for critical incidents.
- **Resource Optimization:** Service Desk staff spend less time manually sorting tickets and more time resolving user issues.

Ideation Output

Selected Concept: Automate incoming Incident classification, priority setting, and assignment group routing through ServiceNow Flow Designer rules triggered on record creation.

Phase 2: Requirement Analysis

Purpose

Translate the project idea into explicit functional and non-functional requirements to configure in ServiceNow.

Functional Requirements

ID	Requirement	Implementation Strategy
FR-01	The flow must trigger automatically when a new Incident record is created.	Flow Designer trigger on Table: incident (Created).
FR-02	The flow must evaluate incoming text or keywords in Short description.	Flow Designer Logic / IF-ELSE Decision Conditions.
FR-03	Category & Subcategory must be set based on ticket classification rules.	Action: Update Record → Set Category & Subcategory.
FR-04	The flow must dynamically update the Assignment group (e.g., Network, Database, Software).	Action: Update Record → Set Assignment Group based on rule match.

FR-05	High-severity keywords must elevate Urgency and Impact.	Action: Update Record → Set Urgency=1 (High), Impact=1 (High).
FR-06	Work Notes must record that auto-classification was performed.	Action: Update Record → Add Work Note: <i>"Auto-classified via Flow Designer"</i> .

Non-Functional Expectations

- **Performance:** Classification logic must execute within seconds of record creation.
- **Reliability:** Fallback logic must assign unclassified tickets to a default *Service Desk* group if no rule matches.
- **Maintainability:** Flow logic should be easily extendable for new classification rules without code modification.

Requirement-Analysis Acceptance Points

- An incident created via portal/email triggers the flow.
- Keywords like *"Network outage"* set Group = **Network**, Category = **Network**.
- Keywords like *"Database error"* set Group = **Database**, Category = **Database**.
- Unmatched incidents default to Group = **Service Desk**.
- System logs show the automated updates in ticket audit trails.

Phase 3: Project Design

Purpose

Architect the ServiceNow components, execution steps, conditional trees, and verification path.

High-Level Architecture

User / Portal / Email → Incident Created
 → Flow Designer Trigger → Classification Rules Engine
 → begin{cases} Match: Network → Assign to Network Group
 || Match: Database → Assign to DB Group
 || No Match → Assign to Service Desk
 end{cases}

Flow Design Specifications

Design Element	Configured Value
Flow Name	Auto Ticket Classification Flow
Application	Global
Run User	System User
Trigger	Created → Table: Incident
Condition 1	Short Description containing "Network" or "VPN" → Category: Network Group: Network

Condition 2	Short Description containing <i>"Database"</i> or <i>"SQL"</i> \$\rightarrow\$ Category: Database Group: Database
Condition 3	Short Description containing <i>"Password"</i> or <i>"Login"</i> \$\rightarrow\$ Category: Software Group: Service Desk
Fallback	Otherwise \$\rightarrow\$ Category: Inquiry / Help Group: Service Desk
Action	Update Record (Incident Record)

Design Verification Path

1. Open **Service Portal** \$\rightarrow\$ Create Standard Incident.
2. Enter keyword-specific short descriptions (e.g., *"Cannot connect to corporate VPN"*).
3. Submit Incident and capture Incident Number (e.g., INC0010001).
4. Review Incident record fields: **Category**, **Subcategory**, **Assignment Group**, and **Work Notes**.

Phase 4: Project Planning

Purpose

Organize team responsibilities, execution phases, and completion checkpoints for a **4-member student team**.

Work Allocation (4-Member Team)

Role	Primary Responsibility	Deliverables
Team Leader (Member 1)	Architecture design, Flow Designer logic, rule tree building, end-to-end integration, overall leadership.	Flow Designer logic configuration, flow activation, team coordination, architecture sign-off.
Member 2	Data modeling, incident field setup, assignment group creation, test data creation.	Category/Subcategory mapping table, Assignment Group definitions, user/group setups.
Member 3	Quality Assurance (QA), execution of test scenarios, defect logging, SLA validation.	Test plan, test cases (TC-01 to TC-08), defect reports, execution matrix.
Member 4	Evidence gathering, documentation,	Screenshots checklist, final PDF project

	screenshots, final presentation (PPT) deck preparation.	document, project presentation slides.
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Implementation Schedule

Stage	Planned Work	Completion Evidence
1. Setup	Configure Assignment Groups (Network, Database, Software, Service Desk) in ServiceNow.	Group records verified.
2. Flow Build	Create Auto Ticket Classification Flow in Flow Designer; set trigger on Incident table.	Draft flow saved.
3. Logic Configuration	Add conditional branch logic (IF/ELSE IF) based on Short Description attributes.	Logic tree built.
4. Actions Setup	Configure Update Record actions to set Category, Priority, and	Action parameters set.

	Assignment Group.	
5. Activation	Test draft flow, fix syntax issues, save, and Activate flow.	Active flow state.
6. Testing	Execute test scenarios across all keyword types and fallbacks.	Test execution evidence.
7. Documentation	Compile system screenshots, architecture diagrams, and test results.	Final report & PPT.

Phase 5: Project Development

Purpose

Step-by-step technical implementation in ServiceNow Flow Designer.

Development Activity 1: Group & Field Configuration

1. Navigate to **User Administration** $\rightarrow$ **Groups**.
2. Verify / Create target groups: Service Desk, Network, Database, and Software.

Development Activity 2: Flow Creation

1. Open **Flow Designer** (Process Automation $\rightarrow$ Flow Designer).
2. Click **New** $\rightarrow$ **Flow**.
3. Name: Auto Ticket Classification Flow.
4. Application: Global.
5. Run As: System User. Click **Submit**.

Development Activity 3: Trigger Configuration

1. Click **Add a Trigger**.
2. Select **Created** under Record operations.
3. Table: **Incident** [incident].
4. Conditions: Active is True.
5. Click **Done**.

Development Activity 4: Conditional Logic & Action Configuration

1. **Branch 1 (Network Logic):**
 - Click **Add an Action/Flow Logic** $\rightarrow$ **Flow Logic** $\rightarrow$ **If**.
 - Condition: Trigger - Record > Short Description **contains** "Network" OR

"VPN".

- Under IF branch: Add Action **Update Record**.
- Select Trigger - Record as Incident record.
- Fields: Category = Network, Assignment Group = Network.

2. **Branch 2 (Database Logic):**

- Add **Else If** condition.
- Condition: Short Description **contains** "Database" OR "SQL".
- Action: **Update Record** $\rightarrow$ Category = Database, Assignment Group = Database.

3. **Branch 3 (Fallback Logic):**

- Add **Else** condition.
- Action: **Update Record** $\rightarrow$ Category = Inquiry / Help, Assignment Group = Service Desk.

4. **Audit Logging Action:**

- Action: **Update Record** $\rightarrow$ Work notes = "Ticket automatically classified and assigned via Flow Designer."

Development Activity 5: Save & Activate

1. Click **Save**.
2. Click **Activate** to make the workflow live.

Phase 6: Project Testing

Purpose

Verify that incoming incidents are automatically categorized and routed according to defined conditions.

Test Scenarios & Test Cases

TC ID	Test Action / Description	Expected Result	Pass / Fail
TC-01	Create Incident with Short Description: <i>"Corporate VPN not connecting"</i> .	Category set to Network ; Group assigned to Network .	PASS
TC-02	Create Incident with Short Description: <i>"SQL Database slow query performance"</i> .	Category set to Database ; Group assigned to Database .	PASS
TC-03	Create Incident with Short Description: <i>"General request for information"</i> .	Fallback triggers: Category set to Inquiry , Group to Service Desk .	PASS

TC-04	Verify Work Notes update.	Work Note added: <i>"Ticket automatically classified..."</i>	PASS
TC-05	Check flow execution context under Flow Designer history.	Flow state shows Completed with green checkmarks on matched branch.	PASS

Phase 7: Project Documentation

Recommended Document & Screenshot Checklist

1. **Flow Designer - Flow Properties:** Screenshot of Name, Application (Global), Run As (System User).
2. **Trigger Configuration:** Screenshot of Incident table trigger on creation.
3. **Flow Logic & Actions:** Screenshot of full IF/ELSE IF flow tree in Flow Designer.
4. **Activated Flow:** Screenshot showing active status badge.
5. **Incident Creation Test:** Screenshot of Incident form submission.
6. **Auto-Classified Ticket:** Screenshot of Incident record showing updated Category, Assignment Group, and Work Notes.

Phase 8: Project Demonstration

Live Demonstration Sequence

1. **Introduction:** State the business problem—manual ticket classification causes assignment delays and human error.
2. **Solution Overview:** Introduce ServiceNow Flow Designer as the automated ticket classification engine.
3. **Flow Architecture:** Show the active flow logic and branches inside Flow Designer.
4. **Live Test 1 (Network Ticket):** Create an incident titled *"VPN connection failing for remote user"*. Show automatic classification to **Network Group**.
5. **Live Test 2 (Database Ticket):** Create an incident titled *"Database server timeout error"*. Show automatic assignment to **Database Group**.
6. **Live Test 3 (Fallback Ticket):** Create an incident titled *"Need general support"*. Show fallback routing to **Service Desk**.
7. **Audit & Conclusion:** Show execution logs and Work Notes entries proving zero manual intervention was required.

Demo Success Criteria

- Presenters explain the problem, trigger logic, and conditional branching clearly.
- All live test tickets classify and route to the correct group instantly upon creation.
- The 4-member team demonstrates clear delegation across leadership, configuration, QA testing, and documentation.